

Learning Evidence

Response to Feedback

To show that each learning objective has been achieved, please divide both the summary and the evidence of learning into sections aligned with the learning objectives. You can find the template in Task List > Number Theory > Resources.

As these are images, I cannot organize them perfectly but I arranged them in a close enough order. I will ensure to keep this in my mind and organize things for future tasks.

Basics

Cristoph Paar: Lecture 2: Modular Arithmetic & Historical Ciphers

- * Goal: computation in finite sets
- * the mod / "modulus" operator - like the remainder operator
- * Let $a, r, m \in \mathbb{Z}$, and $m > 0$,
we write,

$$a \equiv r \pmod{m}$$

(if m divides $(a-r)$, ie, $m \mid (a-r)$)

this is read as
"m divides (a-r)"

- * examples $\Rightarrow$

$$\begin{aligned} a &= 13 & 13 &= 4 \pmod{9} \\ m &= 9 & \Rightarrow & a - r = (13 - 4) = 9 \\ r &= ? \end{aligned}$$

- * computation of the remainder,

* given: $a, m \in \mathbb{Z} \Rightarrow a = qm + r$

quotient
remainder

- * example $\Rightarrow$

$$\begin{aligned} a &= 42 & \Rightarrow & 42 = q \times 9 + r & \Rightarrow & q = 4 \\ m &= 9 & & & & r = 6 \end{aligned}$$

but also,

$$42 = 3 \times 9 + 15 \Rightarrow r = 15$$

$$\text{check} \Rightarrow (42 - 15) = 27, 9 \mid 27 \checkmark$$

but also

$$42 = 5 \times 9 + (-3) \Rightarrow r = -3$$

$$\text{check} \Rightarrow (42 - (-3)) = 45, 9 \mid 45 \checkmark$$

$$\begin{aligned} \text{check} &\Rightarrow 42 - 6 = 36 \\ &9 \mid 36 \checkmark \end{aligned}$$

$\therefore$ the remainder
is always not
unique.

* Equivalence Classes

* Examples $\Rightarrow$ when $a = 12, m = 5,$

$$12 \equiv 2 \pmod{5} \Rightarrow \text{check } 5 \mid (12-2) \checkmark$$

$$12 \equiv 7 \pmod{5} \Rightarrow \text{check } 5 \mid (12-7) \checkmark$$

$$12 \equiv -3 \pmod{5} \Rightarrow \text{check } 5 \mid (12-(-3)) \checkmark$$

so, the set $\{ \dots, -8, -3, 2, 7, 12, 17, \dots \}$

forms an "equivalence class" for modulo 5. All members of the class behave equivalently to modulo 5.

* Lets look at all equivalence classes for modulo 5,

$$\{ \dots, -10, -5, 0, 5, 10, \dots \} - A$$

$$\{ \dots, -9, -4, 1, 6, 11, \dots \} - B$$

$$\{ \dots, -8, -3, 2, 7, 12, \dots \} - C$$

$$\{ \dots, -7, -2, 3, 8, 13, \dots \} - D$$

$$\{ \dots, -6, -1, 4, 9, 14, \dots \} - E$$

from these two lists

$$D = B - D$$

* Lets solve $13 \times 16 - 8 = 208 - 8 = 200 \equiv 0 \pmod{5}$

* Similarly, $3 \times 1 - 3 = 3 - 3 = 0 \pmod{5}$ mod 5,
or steps of 5

$$8 \times 6 - (-7) = 48 + 7 = 55 \equiv 0 \pmod{5}$$

* Important Application

$$E \Rightarrow 3^8 \pmod{7} \equiv ?$$

$\Leftarrow$ exponentiation is what happens in most of cryptography. but the numbers are very large

* 1st method $\Rightarrow 3^8 = 6561 = 2 \pmod{7}$

* 2nd method $\Rightarrow 3^8 = 3^4 \times 3^4 = 81 \times 81$

equivalent of picking from
an equivalent class $\longrightarrow$

$$\begin{array}{ccc} \downarrow & 81 & \downarrow \pmod{7} \\ 4 & \times & 4 = 16 \end{array}$$

basically the
shifting around
we did
before

* Rings: An algebraic view (on modular arithmetic)

* definition $\Rightarrow$ the "integer ring" $\mathbb{Z}_m$ consists of

1. The set $\mathbb{Z}_m = \{0, 1, \dots, m-1\}$

2. Two operators "+" and "*", such that for all $a, b, c \in \mathbb{Z}_m$,

(i) $a + b = c \pmod{m}$

(ii) $a \cdot b \equiv d \pmod{m}$

* we can add, multiply any two numbers and the result is always in the ring. A ring is said to be closed.

* addition and multiplication are associative,

$$a + (b + c) = (a + b) + c$$

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

* there is the neutral element 0 with respect to addition, i.e., for every element $a \in \mathbb{Z}_m$, it holds $a + 0 = a \pmod{m}$.

* for any element 'a' in the ring, there is always the negative element, such that $a + (-a) = 0 \pmod{m}$.

(additive inverse always exists).

* there is a neutral element 1 with respect to multiplication, for every element $a \in \mathbb{Z}_m$, it holds $a \times 1 = a \pmod{m}$.

- the multiplication inverse exists only for some, but not for all elements. Let $a \in \mathbb{Z}$, the inverse a^{-1} is defined such that

$$a \cdot a^{-1} = 1 \pmod{m}$$

if inverse exists for a , we can divide by this element

since $\boxed{b \mid a = b \cdot a^{-1} \pmod{m}}$

- for multiplicative inverse,

let $m=9$, $a=2$, $a^{-1}=?$, $\mathbb{Z}_9 = \{0, 1, 2, 3, 4, \dots\}$

$$\Rightarrow 2 \cdot 2^{-1} = 1 \pmod{9}$$

but $2^{-1} = \frac{1}{2} = 0.5$, is not an integer, $\notin \mathbb{Z}_m$,

so, we are looking for

$$2 \cdot _ = 1 \pmod{9}$$

$$2 \times 5 \equiv 1 \pmod{9}$$

$$2^{-1} \equiv 5 \pmod{9}$$

$$6 \times _ \equiv 1 \pmod{9}$$

$$\gcd(6, 9) = 3 \neq 1$$

$\therefore$ the inverse does not exist

1:14:08

Advanced Number Theory Introduction

①

$$3 = 52 \bmod 7 \Rightarrow 57 \div 7 = 7 \text{ r } 3 \quad (7 \times 7 = 49)$$

* euclidean theory $\Rightarrow \gcd(a, b)$

$$b - a \times m = r_1$$

$$a - r_1 \times m_1 = r_2$$

$$r_1 - r_2 \times m_2 = r_3$$

....

$$\text{rem}(b \div a) = r_1$$

$$\text{rem}(a \div r_1) = r_2$$

$$\text{rem}(r_1 \div r_2) = r_3$$

....

do until $r_{i+1} = 0$, then r_i is the gcd.

$$\begin{aligned} \gcd(16, 36) &\Rightarrow 36 - 16 \times 2 = 4 \\ 16 - 4 \times 4 &= 0 \end{aligned} \quad \therefore \text{gcd is } 4 //$$

* multiplication & addition are associative

$$\hookrightarrow 12 \equiv 3 \pmod{9}$$

$$22 \equiv 4 \pmod{9}$$

$$\left. \begin{aligned} 12 + 22 &= 34 \\ 3 + 4 &= 7 \end{aligned} \right\} \begin{aligned} 34 &\equiv 7 \pmod{9} \\ (34 &= 27 + 7) \end{aligned}$$

$$\left. \begin{aligned} 12 \times 22 &= 244 \\ 3 \times 4 &= 12 \end{aligned} \right\} \begin{aligned} 244 &\equiv 12 \pmod{9} \\ &\equiv 3 \pmod{9} \end{aligned}$$

$$(244 = 243 + 1)$$

* division in modular arithmetic is tricky

Obtain the Bezouts coefficients of a pair of numbers.

* exercise $a \equiv c \pmod{p}$, $a \div m \equiv c \div m \pmod{p}$

is this true?

consider $12 \equiv 3 \pmod{9}$, $22 \equiv 4 \pmod{9}$,

$$12 \times 22 \equiv 12 \pmod{9},$$

but it doesn't mean $22 \equiv 1 \pmod{9}$.

* Bezout's law/lemma / extended euclidean algorithm.

$$* \gcd(65, 30) = 5$$

$$\Rightarrow 65 - 30 \times 2 = 5$$

$$30 - 5 \times 6 = 0$$

$\therefore$ gcd is 5

using the bezout's ~~lemma~~ lemma $\Rightarrow$

$$5 = \boxed{1} \times 65 + \boxed{-2} \times 30$$

$$* \text{example 1} \Rightarrow \gcd(182, 35) = 7$$

$$182 - 35 \times 5 = 7$$

$$35 - 7 \times 5 = 0$$

$$\therefore \gcd(182, 35) = 7$$

bezout's coeffs $\Rightarrow$

$$7 = 182 \times \boxed{1} + \boxed{-5} \times 35$$

Another simple example: $\gcd(182, 35)$:

1. First apply the Euclidean algorithm to find the gcd.
2. Then find the Bézout coefficients.

$$\textcircled{1} \quad 182 - 35 \times 5 = 7$$
$$35 - 7 \times 5 = 0$$

$$\textcircled{2} \quad \underline{1} \times 182 + \underline{-5} \times 35 = 7$$



What about when the Euclidean algorithm takes more than one step? Can you work out how to find the coefficients?

Let us now look at **18** and **48**:

1. Apply the Euclidean algorithm.
2. Rearrange the equations.
3. Sub-in the first equation into the second one.

$$\textcircled{1} \quad \gcd(18, 48) \Rightarrow 48 - 18 \times 2 = 12 \quad \Rightarrow = 6 //$$
$$18 - 12 \times 1 = 6$$
$$12 - 6 \times 2 = 0$$

$$\textcircled{2} \quad \underline{3} \times 18 + \underline{-1} \times 48 = 6$$

$$\begin{aligned} 6 &= 18 - \cancel{12} \\ \cancel{12} &= 48 - 18 \times 2 \\ 6 &= 18 - (48 - 18 \times 2) \\ &= 3 \times 18 - 48 \end{aligned}$$



Another example: 65 and 351:

1. Apply the Euclidean algorithm.
2. Rearrange the equations.
3. Sub-in the first equation into the second one.

Erratum: At 2:43, I forgot to multiply by 2 when simplifying the equation - the coefficients should be 11 and 2, not 6 and -1. Thus, the correct answer should be $13 = 65 - (351 - 65 \times 5) \times 2 = 11 \times 65 - 2 \times 351$.

$$\textcircled{1} \quad 351 - 65 \times 5 = 26$$

$$65 - 26 \times 2 = \underline{13} \quad \gcd(65, 351) = 13$$

$$26 - 13 \times 2 = 0$$

$$\textcircled{2} \quad 13 = 65 - 26 \times 2$$

$$\begin{aligned} \textcircled{3} \quad 13 &= 65 - (351 - 65 \times 5) \times 2 \\ &= 65 - (2 \times 351) + 10 \times 65 \end{aligned}$$



Exercise 1.1.1. Using the Euclidean algorithm, find the gcd and the Bézout coefficients of the following pairs of numbers: (120, 96), (36, 69), (63, 77), (105, 85), (92, 112).

[Solutions](#)

$$\textcircled{1} \quad \gcd(120, 96) \Rightarrow$$

$$120 - 96 \times 1 = \underline{24} \quad \therefore \gcd(96, 120) = 24$$

$$96 - \underline{24} \times 4 = 0$$

$$\underline{-1} \times 96 + \underline{1} \times 120 = 24$$

or I could have also rearranged

$$\textcircled{2} \gcd(36, 69) \Rightarrow$$

$$69 - 36 \times 1 = 33$$

$$36 - 33 \times 1 = \underline{3}$$

$$33 - \underline{3} \times 11 = 0$$

$$\therefore \gcd(36, 69) = 3$$

$$36 - 33 \times 1 = 3$$

$$36 - (69 - 36 \times 1) \times 1 = 3$$

$$36 - 69 + 36 = 3$$

$$2 \times 36 - 69 = 3 //$$

$$\textcircled{5} 63, 77$$

$$77 - 63 \times 1 = 14$$

$$63 - 14 \times 4 = \underline{7}$$

$$14 - 7 \times 2 = 0$$

$$\therefore \gcd(63, 77) = 7$$

$$7 = 63 - (77 - 63 \times 1) \times 4$$

$$7 = 63 - (4 \times 77) + 4 \times 63$$

$$7 = 5 \times 63 - 4 \times 77$$

$$\textcircled{6} \ 105, 85$$

$$105 - 85 \times 1 = 20$$

$$85 - 20 \times 4 = \underline{5} \quad \therefore \gcd(105, 85) = 5$$

$$20 - 5 \times 4 = \underline{0}$$

$$5 = 85 - 4 \cdot (105 - 85 \times 1)$$

$$5 = 85 - 4 \times 105 + 4 \times 85$$

$$5 = 5 \times 85 - 4 \times 105$$

$$\textcircled{7} \ 92, 112$$

$$112 - 92 \times 1 = 20$$

$$92 - 20 \times 4 = 12$$

$$20 - 12 \times 1 = 8$$

$$12 - 8 \times 1 = \underline{4}$$

$$8 - 4 \times 2 = \underline{0}$$

$$\therefore \gcd(92, 112) = 4$$

$$4 = 12 - 8 \times 1$$

$$= 12 - (20 - 12 \times 1)$$

$$= 2 \times 12 - 20 \times 1$$

$$= 2(92 - 4 \times 20) - 1 \times 20$$

$$= 2 \times 92 - 8 \times 20 - 1 \times 20$$

$$= 2 \times 92 - 9 \times 20$$

$$= 2 \times 92 - 9(112 - 92 \times 1)$$

$$= 2 \times 92 - 9 \times 112 + 9 \times 92$$

$$= 11 \times 92 - 9 \times 112$$

Apply the extended Euclidean algorithm for computing the modular inverses.

For example, to find the inverse of $3 \bmod 17$:

apply the extended algorithm $\Rightarrow$

	coeff. corresponding to 17 $\Rightarrow$	coeff. corresponding to 3 $\Rightarrow$
$17 - 3 \times 5 = 2$	$\xrightarrow{1} 1 - 0 \times 5 = 1$	$\xrightarrow{2} 0 - 1 \times 5 = -5$
$3 - 2 \times 1 = 1$	$\xrightarrow{3} 0 - 1 \times 1 = -1$	$\xrightarrow{4} 1 - 1(-5) = 6$
$2 - 1 \times 2 = 0$		

Find the inverse of $7 \bmod 9$

$$9 - 7 \times 1 = 2$$

$$7 - 2 \times 3 = 1$$

$$3 - 1 \times 3 = 0$$

$\therefore$ inverse exists

(from 9)

$1 - 7 \times 0 = 1$
$0 - 1 \times 3 = -3$

$\uparrow$ necessary

(from 7)

$$0 - 1 \times 1 = -1$$

$$1 - (-1) \times 3 = 4$$

this is all we need

$\therefore 4$ is the inverse

let's verify $\Rightarrow$

$$4 \times 7 = 28 = 27 + 1$$

$$4 \times 7 \equiv 1 \pmod{9}$$

can technically stop here

inverse of $3 \bmod 6$.

$$6 - 3 \times 2 = 0$$

$\gcd(3, 6) = 3 \neq 1 \therefore$ no inverse exists

Example: Find the inverse of 35 mod 103.

$$\begin{array}{l|l} 103 - 35 \times 2 = 33 & \text{(35)} \\ 35 - 33 \times 1 = 2 & 0 - 1 \times 2 = -2 \\ 33 - 2 \times 16 = 1 & 1 - (-2) \cdot 1 = 3 \\ & -2 - (3) \times 16 = \underline{-50} \end{array}$$

modular inverse

$\therefore -50$ is the modular inverse

anything congruent to (-50) is also a modular inverse

so, eg $\Rightarrow 53$ $(-50 + 103)$ is also a modular inverse

$$\begin{aligned} 35 \times 53 &\equiv 1885 = 103 \times 18 + 1 \\ &\equiv 1 \pmod{103} \end{aligned}$$

Solve for x : $7x \equiv 5 \pmod{9}$.

We know that $4 \times 7 \equiv 1 \pmod{9}$

$$x \times 7 \equiv 5 \pmod{9}$$

$\times 4 \Rightarrow$

$$\begin{aligned} x \times 7 \times 4 &\equiv 5 \times 4 \pmod{9} \\ \underbrace{\quad}_{1 \pmod{9}} & \end{aligned}$$

$$x \equiv 20 \pmod{9} \equiv 2 \pmod{9}$$

linear equations with mod- arithmetic

If p is a prime number, then any linear equation $ax \equiv b \pmod{p}$ has a solution, except if $a \equiv 0 \pmod{p}$.

Proof: If $a \not\equiv 0 \pmod{p}$, then a is not a multiple of p , and so $\gcd(a, p) = 1$.

PART A Find the modular inverses, if they exist, of the following numbers.

(a) 25 mod 111,

(25)

$$111 - 25 \times 4 = 11$$

$$25 - 11 \times 2 = 3$$

$$11 - 3 \times 3 = 2$$

$$3 - 2 \times 1 = 1$$

$$\gcd(25, 111) = 1$$

$\therefore$ they has a modular inverse

$$0 - 1 \times 4 = -4$$

$$1 - (-4 \times 2) = 9$$

$$-4 - (9 \times 3) = -31$$

$$9 - -31 \times 1 = 40$$

$\nearrow$
inverse

$$40 \times 25 \equiv 1000 \equiv 1 \pmod{111}$$

(b) 75 mod 116,

(75)

$$116 - 75 \times 1 = 41$$

$$75 - 41 \times 1 = 34$$

$$41 - 34 \times 1 = 7$$

$$34 - 7 \times 4 = 6$$

$$7 - 6 \times 1 = 1$$

$\therefore$ inverse exists
as $\gcd$ is 1

$$0 - 1 \times 1 = -1$$

$$1 - (-1 \times 1) = 2$$

$$-1 - 2 \times 1 = -3$$

$$2 - -3 \times 4 = 14$$

$$-3 - 14 \times 1 = -17$$

$\nearrow$
inverse

$$75 \times \underline{-17} + 116 \times \underline{11} = 1$$

$$116 - 17 = 99$$

$$99 \times 75 = 7425 \equiv 1 \pmod{116}$$

$$\frac{75 \times 17}{116} = 10.9$$

(c) 29 mod 77,

$$77 - 29 \times 2 = 19$$

$$29 - 19 \times 1 = 10$$

$$19 - 10 \times 1 = 9$$

$$10 - 9 \times 1 = 1$$

29

$$0 - 1 \times 2 = -2$$

$$1 - (-2) \cdot 1 = 3$$

$$-2 - 3 \cdot 1 = -5$$

$$3 - -5 \cdot 1 = \underline{8} \quad \swarrow \text{inverse}$$

$$8 \times 29 = 232 \equiv 1 \pmod{77}$$

(d) 125 mod 104,

lets first reduce 125,

$$125 = 1 \times 104 + 21$$

$$125 \bmod 104 = 21$$

$$125 = 21 \pmod{104}$$

$$104 - 21 \times 4 = 20$$

$$21 - 20 \times 1 = 1$$

21

$$0 - 4 = -4$$

$$1 + 4 = 5$$

$$125 \times 5 = 625 \equiv 1 \pmod{104}$$

(e) 87 mod 28.

lets reduce it,

$$87 = 28 \times 3 + 3$$

$$87 \bmod 28 = 3$$

$$87 = 3 \pmod{28}$$

$$28 - 3 \times 9 = 1$$

3

$$0 - 1 \cdot 9 = -9$$

inverse

↓

19 //

$$87 \times 19 = 1653 \equiv 1 \pmod{28}$$

Solve linear equations in modular arithmetic.

PART B Use the results from **PART A** to solve the following linear equations.

(f) $25x \equiv 2 \pmod{111}$. (Solution: $x \equiv 80 \pmod{111}$)

$$40 \times 25 \equiv 1000 \equiv 1 \pmod{111} \text{ --- ①}$$

$$25n \equiv 2 \pmod{111}$$

$$\times 40 \Rightarrow$$

$$40 \times 25 \equiv 40 \times 2 \pmod{111}$$

$$40 \cdot 25 \equiv 80 \pmod{111}$$

$$\text{from ①} \Rightarrow$$

$$n \equiv 80 \pmod{111}$$

(g) $75x \equiv 4 \pmod{116}$. (Solution: $x \equiv 48 \pmod{116}$)

$$99 \times 75 = 7425 \equiv 1 \pmod{116} \text{ --- ①}$$

$$75n \equiv 4 \pmod{116}$$

$$99 \times 75n \equiv 99 \times 4 \pmod{116}$$

$$99 \cdot 75n \equiv 396 \pmod{116}$$

$$\text{from ①} \Rightarrow$$

$$n \equiv 396 \pmod{116}$$

$$396 = 116 \times 3 + 48$$

$$396 \equiv 48 \pmod{116}$$

$$n \equiv 48 \pmod{116}$$

(h) $29x \equiv 8 \pmod{77}$. (Solution: $x \equiv 64 \pmod{77}$)

$$8 \times 29 = 232 \equiv 1 \pmod{77}$$

$$29n \equiv 8 \pmod{77}$$

$\times 8 \Rightarrow$

$$29 \times 8 n \equiv 8 \cdot 8 \pmod{77}$$

$$8 \cdot 29n \equiv 64 \pmod{77}$$

$$1n \equiv 64 \pmod{77}$$

$$n \equiv 64 \pmod{77}$$

(i) $125x \equiv 3 \pmod{104}$. (Solution: $x \equiv 15 \pmod{104}$)

$$125 \times 5 = 625 \equiv 1 \pmod{104} \text{ --- ①}$$

$$125n \equiv 3 \pmod{104}$$

$\times 5 \Rightarrow$

$$125 \cdot 5 n \equiv 3 \cdot 5 \pmod{104}$$

from ① $\Rightarrow$

$$1n \equiv 3 \cdot 5 \pmod{104}$$

$$n \equiv 15 \pmod{104}$$

(j) $87x \equiv 5 \pmod{28}$. (Solution: $x \equiv 11 \pmod{28}$)

$$87 \times 19 = 1653 \equiv 1 \pmod{28} \text{ --- ①}$$

$$87n \equiv 5 \pmod{28}$$

$\times 19 \Rightarrow$

$$87 \cdot 19 n \equiv 5 \cdot 19 \pmod{28}$$

$$1n \equiv 5 \cdot 19 \pmod{28}$$

$$1n \equiv 11 \pmod{28}$$

$$n \equiv 11 \pmod{28}$$

$$87 \equiv 3 \pmod{28}$$

$$5 \times 19 = 95$$

$$28 \times 3 = \frac{84}{11} -$$

Apply the Chinese remainder theorem to solve systems of two linear congruences.

Solve for x : $x \equiv 3 \pmod{7}$ and $x \equiv 5 \pmod{15}$.

Chinese remainder theorem

$$\gcd(7, 15) \Rightarrow 15 - 7 \times 2 = 1 \Rightarrow 1$$

(7)

$$0 - 1 \times 2 = -2$$

(15)

$$1 - 0 = 1$$

$$\Rightarrow 1 = 15 \times \underline{1} + 7 \times \underline{-2}$$

$$x \equiv bms + ant \pmod{m \cdot n}$$

$$x \equiv 5 \cdot 7 \cdot -2 + 3 \cdot 15 \cdot 1 \pmod{7 \cdot 15}$$

$$x \equiv -70 + 45 \pmod{105}$$

$$x \equiv -25 \pmod{105}$$

$$x \equiv 80 \pmod{105}$$

Exercise 1.3.2. Solve for x : $x \equiv 5 \pmod{7}$, $x \equiv 7 \pmod{11}$, $x \equiv 3 \pmod{13}$.

extended euclidean algorithm $(7, 11) \Rightarrow$

$$\begin{array}{l|l|l} (7) & (11) & \\ \hline 11 - 7 \times 1 = 4 & 0 - 1 \cdot 1 = -1 & 1 - 0 = 1 \\ 7 - 4 \times 1 = 3 & 1 - 1 \cdot 1 = 2 & 0 - 1 \cdot 1 = -1 \\ 4 - 3 \times 1 = 1 & -1 - 2 \cdot 1 = -3 & 1 - 1 \cdot 1 = 2 \end{array}$$

$$\therefore 1 = 7 \times \underline{-3} + 11 \times \underline{2}$$

$$x \equiv bms + ant \pmod{m \cdot n}$$

$$x \equiv 7 \cdot 7 \cdot -3 + 5 \cdot 11 \cdot 2 \pmod{7 \cdot 11}$$

$$x \equiv -37 \pmod{77}$$

$$x \equiv \underline{40} \pmod{77}$$

$$\Rightarrow x \equiv \underline{3} \pmod{13}$$

$$\begin{array}{l|l|l} (77) & (13) & \\ \hline 77 - 13 \times 5 = 12 & 1 - 0 = 1 & 0 - 1 \times 5 = -5 \\ 13 - 12 \times 1 = 1 & 0 - 1 \cdot 1 = -1 & 1 - -5 \cdot 1 = 6 \end{array}$$

$$x \equiv bms + ant \pmod{m \cdot n}$$

$$\equiv 3 \cdot 77 \cdot (-1) + 40 \cdot 13 \cdot 6 \pmod{77 \times 13}$$

$$\equiv 2889 \pmod{1001}$$

$$x \equiv 887 \pmod{1001}$$

$$2889 - 1001 \times 2 = 887$$

Apply the Chinese remainder theorem to solve systems of two linear congruences.

$$\begin{matrix} a & m & b & n \\ \swarrow & \searrow & \swarrow & \searrow \end{matrix}$$

Exercise 1.3.3. Solve for x : $x \equiv 2 \pmod{5}$, $x \equiv 4 \pmod{7}$, $x \equiv 1 \pmod{9}$.

$$(5, 7) \Rightarrow \left| \begin{array}{c} \underline{5} \\ 7 - 5 \times 1 = 2 \\ 5 - 2 \times 2 = 1 \end{array} \right| \left| \begin{array}{c} \underline{5} \\ 0 - 1 \cdot 1 = -1 \\ 1 - -1 \cdot 2 = 3 \\ \quad \quad \quad \uparrow \\ \quad \quad \quad 5 \end{array} \right| \left| \begin{array}{c} \underline{7} \\ 1 - 0 = 1 \\ 0 - 1 \cdot 2 = -2 \\ \quad \quad \quad \uparrow \\ \quad \quad \quad 7 \end{array} \right|$$

$$x \equiv bms + ant \pmod{m \cdot n}$$

$$x \equiv 4 \cdot 5 \cdot 3 + 2 \cdot 7 \cdot -2 \pmod{5 \cdot 7}$$

$$x \equiv \underline{a} \underline{32} \pmod{\underline{35} \underline{m}}$$

$$\Rightarrow x \equiv \underline{b} \underline{1} \pmod{\underline{9} \underline{q}}$$

$$(9, 35) \Rightarrow \left| \begin{array}{c} \underline{9} \\ 35 - 9 \times 3 = 8 \\ 9 - 8 \times 1 = 1 \end{array} \right| \left| \begin{array}{c} \underline{9} \\ 0 - 1 \cdot 3 = -3 \\ 1 + 3 = 4 \\ \quad \quad \quad \uparrow \\ \quad \quad \quad 6 \end{array} \right| \left| \begin{array}{c} \underline{35} \\ 1 - 0 = 1 \\ 0 - 1 = -1 \\ \quad \quad \quad \uparrow \\ \quad \quad \quad 5 \end{array} \right|$$

$$x \equiv bms + ant \pmod{m \cdot n}$$

$$\equiv 1 \cdot 35 \cdot -1 + 32 \cdot 9 \cdot 4 \pmod{35 \cdot 9}$$

$$\equiv 1117 \pmod{315}$$

$$\equiv 172 \pmod{315}$$

$$\left. \begin{array}{l} \frac{1117}{315} = 3.5 \\ 1117 - 315 \times 3 = 172 \end{array} \right\}$$

Try your own: pick a number p (make sure it's prime!) and a number a (make sure it's not divisible by p !) and verify that $a^p \equiv a \pmod{p}$.

$$p=7, a=12$$

$$\begin{aligned} a^p &= 12^7 = 35831808 \\ &= 5118829 \times 7 + 5 \end{aligned}$$

$$\left. \begin{array}{l} 12^7 \equiv 5 \pmod{7} \\ 12 \equiv 5 \pmod{7} \end{array} \right\} \therefore 12^7 \equiv 12 \pmod{7}.$$

To prepare for the proof, let's get back to our example:

$$p=7, a=12$$

$$12 \times 1 = 5 \pmod{7}$$

$$12 \times 2 = 3 \pmod{7}$$

$$12 \times 3 = 1 \pmod{7}$$

$$12 \times 4 = 6 \pmod{7}$$

$$12 \times 5 = 4 \pmod{7}$$

$$12 \times 6 = 2 \pmod{7}$$

as seen here,
we got $\underbrace{1 \rightarrow 6}_{\uparrow} \leftarrow (6 \text{ is } 7-1)$

all the numbers
as the result.

now we multiply RHS & LHS

$$\underbrace{1 \cdot 12}_5 * \underbrace{2 \cdot 12}_3 * \underbrace{3 \cdot 12}_1 * \underbrace{4 \cdot 12}_6 * \underbrace{5 \cdot 12}_4 * \underbrace{6 \cdot 12}_2 \Rightarrow 12^6 (1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6) \\ \rightarrow (1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6) \pmod{7}$$

$$12^6 \equiv 1 \pmod{7}$$

$$12^7 \equiv 12 \pmod{7}$$

prime. $\nwarrow$
(we can simplify)

PROOF

What did we do?

1. Take the remainder of $k \cdot a$ in the division by p .
2. Observe that each remainder appears exactly once (ie, we get the same numbers from 1 to $p-1$)
3. Multiply the left-hand-sides and right-hand sides
4. Simplify

$$a^p \equiv a \pmod{p} \quad - p \text{ is prime, } a \text{ is } \neq \text{ a multiple of } p$$

$$1. \quad k \cdot a = r_k \pmod{p} \quad - \text{ for every } k \text{ from } 1 \text{ to } p-1$$

$$2. \quad \text{we don't get } r_k = 0,$$

because $(k \cdot a - r_k)$ is a multiple of p

$k \cdot a$ is not a multiple of p (as p is prime)

$k \cdot a$ " " " " " (as its smaller than p)

why $r_k \neq r_j$ if $k \neq j$?

$$\underbrace{[(k-j) \cdot a - (r_k - r_j)]}_{\text{not a multiple of } p}$$

r_k 's take all values between 1 & $p-1$

3. multiply on both sides

$$\prod_{k=1}^{p-1} k \cdot a \equiv \prod_{k=1}^{p-1} r_k \pmod{p}$$

$$a^{p-1} \prod_{k=1}^{p-1} k \equiv \prod_{k=1}^{p-1} k \pmod{p}$$

$$4 - a^{p-1} \equiv 1 \pmod{p}$$

$$a^p \equiv a \pmod{p}$$

Apply Fermats and Eulers theorems to compute high powers in modular arithmetic.

Exercise 1.4.3. Use Euler and Fermat's theorems to determine:

1. $15^{4042} \bmod 41$

↓
is prime

Fermat's theorem $\Rightarrow 15^{41} \equiv 15 \bmod 41$ — ①

$$15^{40} \equiv 1 \bmod 41 \text{ — ②}$$

divide 4042 by 40

$$4042 - 101 \times 40 = 2$$

as ① $\Rightarrow 15^{4042} \equiv 15^2 \bmod 41$
 $\equiv 225 \bmod 41$
 $\equiv 14 \times 5 + 20 \bmod 41$
 $\equiv 20 \bmod 41$

2. $7^{47623} \bmod 10$ $\rightarrow$ not prime $\therefore$ use eulers

Euler: $7^{\phi(10)} \equiv 1 \bmod 10$

$$\phi(10) = |\{m \mid \text{relatively prime to } 10\}|$$

$$\begin{array}{ccccccc} 1, 2, 3, 4, 5, 6, 7, 8, 9, \dots \\ \uparrow \quad \uparrow \quad \quad \uparrow \quad \uparrow \end{array}$$

$$\therefore \phi(10) = 4$$

$$7^4 \equiv 1 \bmod 10$$

47620 is divisible by 4

$$\begin{aligned} 7^{47620} &\equiv 1 \bmod 10 \\ &\equiv 7^3 \bmod 10 \\ &\equiv 49 \times 7 \bmod 10 \\ &\equiv 9 \times 7 \bmod 10 \\ &\equiv 63 \bmod 10 \\ &\equiv 3 \bmod 10 \end{aligned}$$

Exercise 1.4.4. Calculate:

1. $11^{27699} \bmod 13$ — prime

$$| a^p \equiv a \pmod{p}$$

$$\gcd(11, 13) \Rightarrow \begin{aligned} 13 - 11 \times 1 &= 2 \Rightarrow 1 \\ 11 - 2 \times 5 &= 1 \end{aligned}$$

$$11^{13-1} \equiv 1 \pmod{13}$$

$$11^{12} \equiv 1 \pmod{13}$$

$$27699 - 12 \times 2308 = 3$$

$$27699 = 12 \times 2308 + 3$$

$$\begin{aligned} \therefore 11^{27699} &= 11^{12 \cdot 2308 + 3} \\ &= (11^{12})^{2308} \times 11^3 \\ &\equiv 1^{2308} \cdot 11^3 \pmod{13} \\ &\equiv 11^3 \pmod{13} \\ &\equiv 5 \pmod{13} // \end{aligned}$$

$$\begin{aligned} | 11^3 &\equiv (-2)^3 \\ &\equiv -8 \\ &\equiv 5 \pmod{13} \end{aligned}$$

2. $2^{30163} \bmod 17$ — prime

$$\text{Fermat's theorem} \Rightarrow \begin{aligned} 2^{17-1} &\equiv 1 \pmod{17} \\ 2^{16} &\equiv 1 \pmod{17} \end{aligned}$$

$$\begin{aligned} | 30163 - 16 \times 1885 &= 3 \\ 30163 &= 16 \cdot 1885 + 3 \end{aligned}$$

$$\begin{aligned} \text{so, } 2^{30163} &= 2^{16(1885) + 3} \\ &= (2^{16})^{1885} \times 2^3 \\ &= 1^{1885} \times 2^3 \pmod{17} \\ &= 2^3 \pmod{17} \\ &= 8 \pmod{17} // \end{aligned}$$

3. $7^{31317} \bmod 10$ — not prime

$$\gcd(7, 10) \Rightarrow 10 - 7 \times 1 = 3 \Rightarrow 1$$

$$7 - 3 \times 2 = 1$$

$$\phi \Rightarrow \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$\uparrow \quad \uparrow \quad \quad \quad \uparrow \quad \quad \uparrow$

$$\therefore \phi(10) = 4$$

$$\text{Euler's Theorem} \Rightarrow 7^4 \equiv 1 \bmod 10$$

$$31317 = 4 \times 7829 + 1$$

$$7^{31317} = 7^{4(7829) + 1}$$

$$= (7^4)^{7829} \times 7$$

$$\equiv 1^{7829} \cdot 7 \bmod 10$$

$$\equiv 7 \bmod 10 //$$

4. $11^{11331} \bmod 16$ — x prime

$$\gcd(11, 16) \Rightarrow 16 - 11 \times 1 = 5 \Rightarrow 1$$

$$11 - 5 \times 2 = 1$$

$$\phi \Rightarrow \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$$

$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8$

$$\Rightarrow \phi(16) = 8$$

$$\therefore 11^8 \equiv 1 \bmod 16$$

$$11^{11331} = 11^{8 \times 1416 + 3}$$

$$= (11^8)^{1416} \times 11^3$$

$$\equiv 1^{1416} \times 11^3 \bmod 16$$

$$\equiv 1331 \bmod 16$$

$$\equiv 3 \bmod 16$$

$$11331 = 8 \times 1416 + 3$$

$$1331 = 16 \times 83 + 3$$

5. $9^{12096} \bmod 20$

$$\gcd \Rightarrow 20 - 1 \times 2 = 2 \Rightarrow 1 \text{ is not prime} \therefore \text{euler}$$

$$9 - 2 \times 4 = 1$$

$$\phi \Rightarrow \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19\}$$

$$\begin{array}{cccccccc} & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{array}$$

$$\therefore \phi(20) = 8$$

$$9^8 \equiv 1 \bmod 20$$

$$12096 = 8 \times 1512$$

$$\begin{aligned} 9^{12096} &= 9^{8 \times 1512} \\ &= (9^8)^{1512} \\ &= 1^{1512} \bmod 20 \\ &= 1 \bmod 20 \end{aligned}$$

$$\therefore 9^{12096} \equiv 1 \bmod 20$$

Exercise 1.4.5. What is the inverse of:

1. $8 \bmod 11$

$$8^{\phi(11)-1} \bmod 11 \quad | \quad \phi(11) = 10$$

$\therefore 8^9$ is the inverse of $8 \bmod 11$

$$8n \equiv 1 \bmod 11$$

$$\Rightarrow 8 \times 2 = 16 \equiv 5 \bmod 11$$

$$8 \times 3 = 24 \equiv 2 \bmod 11$$

$$8 \times 4 = 32 \equiv 10 \bmod 11$$

...

$$8 \times 7 = 56 \equiv 1 \bmod 11$$

$\uparrow$

$$\therefore \text{inverse}(8^9) \equiv 7 \bmod 11$$

$$\begin{array}{l} 11 \\ 22 \\ 33 \\ 44 \\ 55 \leftarrow \end{array}$$

2. $3 \bmod 10$ — not prime

$$\phi(10) \Rightarrow \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$\uparrow \quad \uparrow \quad \quad \uparrow \quad \quad \uparrow$

$$\therefore \phi(10) = 4$$

$$\Rightarrow 3n \equiv 1 \bmod 10$$

$$3 \times 1 = 3, \quad 3 \times 2 = 6, \quad 3 \times 3 = 9$$

$$3 \times 4 = 12 \equiv 2 \bmod 10$$

$$3 \times 5 = 15 \equiv 5 \bmod 10$$

$$3 \times 6 = 18 \equiv 8 \bmod 10$$

$$3 \times 7 = 21 \equiv 1 \bmod 10$$

$$\therefore 3 \times 7 \equiv 1 \bmod 10$$

$$\therefore \text{inverse}(3) = 7 \bmod 10$$

Exercise 1.4.6. Find the last two digits of 11^{402} .

$$11^{402} \bmod 100 \Rightarrow$$

$$\gcd(11, 100) \Rightarrow 100 - 11 \times 9 = 1 \Rightarrow 1$$

$$11^{\phi(100)} \equiv 1 \bmod 100$$

$$\begin{aligned}\phi(100) &= \phi(2^2 \times 5^2) = 100 \times \left(1 - \frac{1}{2}\right) \times \left(1 - \frac{1}{5}\right) \\ &= 100 \times \frac{1}{2} \times \frac{4}{5} \\ &= 40\end{aligned}$$

$$11^{40} \equiv 1 \bmod 100$$

$$\begin{aligned}11^{402} &= 11^{40 \times 10} \times 11^2 \\ &\equiv 1^{10} \times 11^2 \bmod 100 \\ &\equiv 121 \bmod 100 \\ &\equiv 21 \bmod 100\end{aligned}$$

$$402 = 40 \times 10 + 2$$

$\therefore$ the last two digits of 11^{402} are 21